

Predictability of Treasury Bond Returns: Risk Premia or Overreaction?

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Abstract

We examine whether the predictability and business-cycle dependence of excess returns in US Treasuries can be more naturally explained in terms of state-dependent risk premia or a specific cognitive bias (representativeness). We show that the extremely parsimonious cognitive-bias model in Shleifer and Gennaioli (2018) accounts very well for a large number of stylized facts about the predictability of excess returns, and their business-cycle dependence. We also test the risk-premium explanation by looking at the correlation between the payoff of the carry strategy and (several proxies for) consumption. When we do so we find that this correlation either has no explanatory power for returns, or has the wrong sign. We conclude that undue extrapolation in the future of recent information provides a parsimonious, simple and arguably more compelling account of excess returns than a risk-premium explanation.

1 Where Does the Predictability of Treasury Excess Returns Come From?

The predictability of excess returns in Treasuries using ‘carry’ strategies (defined below) has been established since the 1990s. The most common explanation for this profitability points to a (state-dependent) risk premium for assuming duration (level) risk. Is this explanation convincing? Are there simpler mechanisms that can account for the observed predictability as convincingly, or perhaps more so? This is the question addressed in this paper. The answer is in the affirmative, and points to the role played by investors’ overreactions.

The problem can be framed more precisely as follows. Many studies have established that a trading strategy of buying systematically (unconditionally) long-maturity Treasury bonds, and funding this position by selling short-dated bonds provides a positive excess return. (In what follows we call this strategy the unconditional ‘carry’ strategy – see also Equation (1) for a precise definition.) The data in Tab I, which reports descriptive statistics of yearly excess returns

for the unconditional carry strategy for the yield maturities displayed in the top row, show the magnitude of the effect.

Maturity	2	3	4	5	6	7	8	9	10
Average	0.0065	0.0117	0.0161	0.0200	0.0233	0.0262	0.0287	0.0309	0.0328
Avg/Dur	0.0032	0.0039	0.0040	0.0040	0.0039	0.0037	0.0036	0.0034	0.0033
Std Dev	0.0180	0.0327	0.0456	0.0573	0.0685	0.0792	0.0898	0.1002	0.1106
SR	0.3586	0.3569	0.3538	0.3483	0.3404	0.3307	0.3199	0.3083	0.2967

Tab I: Descriptive statistics of yearly excess returns, calculated using Equation (1), for the yield maturities displayed in the top row. Rows two to five show the average excess return, the average excess return divided by duration, the standard deviation of the excess returns, and the associated Sharpe Ratio, *SR*, respectively.

It is also well established that there exist *conditional* carry strategies (Fama and Bliss (1987), Campbell and Shiller (1991), and, more recently, Cochrane and Piazzesi, 2005, Cieslak Povala, 2010) that, by making use of yield-curve risk-predicting factors, can produce substantially higher excess returns than the unconditional strategy. Similarly, conditioning on the state of the economy (Cooper and Priestly (2009) uses the output gap), or on macrofinancial variables (Ludvigson and Ng, 2009) has also been shown to substantially improve the unconditional return predictability.

All these return-predicting factors suggest that information about the shape of the yield curve or about the state of the economy can substantially improve the unconditional returns. See, in this respect, Tab II, which focusses on the business-cycle dependence of the Sharpe ratio of the unconditional and conditional carry strategies (where the conditioning in the table is over the state of the economy). The most noticeable feature is how strongly the Sharpe ratio depends on the phase of the business cycle, ranging from 0.82 in recessions to -0.61 in the second phase of expansions.

	2-year	5-year	10-year
Full Sample	0.20	0.20	0.16
1955-1986	0.04	-0.01	-0.07
1987-2014	0.59	0.56	0.49
Recession	0.82	0.72	0.59
Expansion	0.01	0.06	0.05
<i>1st half Expansion</i>	0.52	0.50	0.45
<i>2nd half Expansion</i>	-0.61	-0.50	-0.48
Tightening Cycles			
1979:Q3-1981:Q2	-1.06	-1.13	-1.23
1993:Q3-1995:Q1	-0.79	-0.86	-0.86
2004:Q2-2006:Q2	-1.52	-0.90	-0.50

Tab II: Sharpe ratios for the excess return ‘carry’ strategy for 2, 5 and 10 years investment horizons applied to US Treasuries during the 1955-2014 period, subdivided i) into different chronological sub-periods, ii) into periods of recessions or expansions, and iii) during tightening cycles.

There is little statistical doubt that return-predicting factors predict, ie, that conditional strategies are more profitable than unconditional ones. But, why is this the case?

The most natural answer from asset pricing theory is that the magnitude of the risk premium associated with assuming duration risk is state dependent. The business-cycle dependence of risk aversion is well documented in the case of equities (see, eg, Baker and Wurgler (2006)), but, as Laborda and Olmo (2013) point out, there is a relative paucity of empirical studies assessing the impact of investor sentiment for explaining and predicting bond risk premia. Laborda and Olmo (2013) make a contribution to filling this gap by using the sentiment index built in Baker and Wurgler (2006) and adding it to the Cochrane-Piazzesi and Ludvigson-Ng factors. They find that the additional sentiment factor adds predictability, but the increase in R^2 is relatively modest. Along similar lines, Baker and Wurgler (2012) study the relationship between sentiment (proxied by their index) and the comovements of government bonds and ‘bond-like’ equity stocks, and find that when sentiment is high bond returns are expected to be positive; in particular they invoke a time-varying investor sentiment that is linked to market risk aversion as one of the explanations for their findings. Fama (1986), Stambaugh (1988), Fama and French (1989), Dahlquist and Hasseltoft (2016) argue along similar lines, and the intuition behind all these results is that ‘bond risk premia [are] tied to a compensation required by the investor for bearing risks related to recessions’.¹

However, investors becoming more or less risk averse in different states of the world cannot by itself explain the findings reported in Tab II, from which it is clear that the unconditionally positive Sharpe ratios displayed in Tab I actually come from averaging strongly positive and strongly negative state-dependent Sharpe ratios. But, as we show below, Sharpe ratios with changing signs imply risk premia and market prices of risk that also change sign. Short of assuming that investors become risk-seeking (not just less risk averse) in some states of the world, the cyclical change in sign of the Sharpe ratio cannot be explained simply in terms of a state-dependent degree of (same-sign) risk aversion.

A change in the sign of the risk compensation commanded by rational, unbiased investors could, of course, be explained even with constant risk aversion if the covariance between the payoffs from Treasuries and some proxy for consumption also changed sign (see the simple set-up in Section 6) – ie, if the asset in question is perceived as providing insurance against consumption shocks. But is this the case? Are the changes in sign of the covariance of the right timing and magnitude to account for the observed changes in the risk premia? Can, in short, the observed state-dependence of the excess returns be explained

¹Laborda and Olmo (2013) , page 208.

in terms of a sign-changing covariance (as a risk-premium explanation predicts should be the case)? We explore this possibility in the body of the paper, but we can anticipate that we find either no relationship or, if significant, a relationship with the opposite sign to what required for a risk-premium / rational-investor explanation to hold.

If an asset-pricing account of the predictability of excess returns leaves some unexplained gaps, what are the main explanatory contenders? Cognitive biases are an obvious candidate. For instance, Bulkley, Harris and Nawosah (2015) decompose excess bond returns into a component related to errors in expectations and a component related to revisions of expectations. When they do so, they find systematic errors both in expectations and in revisions, and explain these errors by invoking two behavioural biases: conservatism and the ‘law of small numbers’ (the latter being the tendency to expect the moments of a population to be reflected in short samples of data, giving rise to unwarranted extrapolation). Conservatism, on the other hand, is reflected in individuals attaching too much weight to their current beliefs and too little to new information. We note that these two behavioural biases ‘pull’ in opposite directions, and can therefore explain, or account for, a great variety of observed behaviours. This lack of parsimony is a common feature of many behavioural-finance accounts of pricing anomalies, and the very richness of candidate biases often strands in the way of ‘crisp’ predictions. In this paper we try to avoid this embarrassment of riches by following the approach proposed in Shleifer and Gennaioli (2018), who only allow for one type of cognitive bias (representativeness, which give rise to extrapolation and overreaction), and provide clearly testable predictions.

We incorporate the insight by Shleifer and Gennaioli (2018) by building a simple monetary model with risk-neutral but overreacting investors. We show that we find a clear indications that the predictability of excess returns, the deviations from the Expectation Hypothesis that Laborda and Olmo set out to explain and the many stylized facts about return-predicting factors can all be readily accounted in terms of overreaction: for instance, the slope predicts excess returns (in magnitude and sign), excess returns display business-cycle variation, and the R^2 of excess return/slope regression is consistent with empirical findings.

We cannot exclude that a state-dependent degree of risk aversion, and a compensation for bearing duration risk may also play a role, but both these mechanisms also require ‘something else’. A judicious use of Occam’s razor suggests that this something else could be the overreaction model by Shleifer and Gennaioli (2018).

In the light of these considerations the paper is organized in four parts.

In the first part we compare and analyze the predictions of future yields and yield changes produced by different popular return-predicting factors on the one hand, and by the expectation hypothesis on the other, in order to understand why the expectation-hypothesis predictions are worse than the predictions from the return-predicting factors. We show that the expectation-hypothesis predictions are biased, and that the nature of the bias naturally points to an overreaction model. At this stage of the analysis, an overreaction explanation naturally offers itself as a candidate worthy of deeper analysis.

In the second part of the paper we therefore explore whether an overreaction model can indeed recover the salient feature of the observed predictability highlighted in the introduction. We do so by using the extremely parsimonious approach by Shleifer and Gennaioli (2018), who only make use of one bias (representativeness) and show how it can explain a wide array of deviations from rational Bayesian updating. We stress that in their set-up (and in the model we build) all the weight of the overreaction explanation rests on the shoulders of one single parameter. We therefore set up a simple negative-feedback model of the actions of the monetary authorities and the decisions of Treasury investors, both in the case when risk-neutral investors are fully rational and unencumbered by any cognitive biases, and when investors are still risk-neutral, but deviate from these standards of rationality in the very simple, robust and parsimonious manner posited by Shleifer and Gennaioli, ie, by overreacting. When we do so, we find that if investors are unbiased there is no return predictability, but that, if they overreact, we can explain a number of salient features of Treasury excess returns, and, *in primis*, the observed state dependence of excess returns on the slope of the yield curve and on the state of the economy (which is monitored and influenced by the central bank in our simple model).

In the third part of the paper we then explore whether it is necessary to add a risk premium explanation to account for the predictability afforded by the return-predicting factors. We do so by testing for the statistical significance of the sign of the correlation between Treasury returns and (some proxies for) consumption growth to explain state-dependent excess returns. We find no evidence that risk premia account for the return predictability. Indeed, when we test for the linear dependence of excess returns or of return-predicting factors on the covariance between Treasury returns and some proxies for consumption growth, we fail to find any statistically significant dependence (if anything, the correlations that we find have the wrong sign!). Admittedly, finding convincing proxies for consumption growth is not easy, but, if proxying for consumption growth is so difficult, arguably a risk-premium explanation of excess returns becomes very difficult to falsify.

Finally, in the fourth part of the paper we provide a rebuttal to the common (and *prima facie* very convincing) argument that points to the upward sloping nature of the time-averaged yield curve to imply that risk premia must be responsible for this feature.

We present our conclusions in Section 9.

2 The Data

For our study we use the US Treasury discount-bond data made available by the Fed and created using the approach in Gurkaynak, Sack and Wright (2007). Since risk premia should grow approximately linearly with maturity, to have a stronger signal we extend the maturity range to ten years (Cochrane and Piazzesi stop at five years). Extending the maturity range restricts somewhat the data availability, that now starts in 1971. However, at this writing we have

at our disposal more than 10 years of extra data than Cochrane and Piazzesi did in 2005.

In order to keep the analysis and intuition as simple and clear as possible, we proxy consumption growth in a very simple way: we create a composite index as a convex combination of four common and plausible consumption proxies, namely the total return from the S&P500 index, GDP growth, real disposable income and one minus the so-called ‘anxious index’ compiled by the Philadelphia Fed. (This is the probability of quarter-over-quarter chain-weighted real GDP growth less than zero for the current quarter (RECESS1) and the following four quarters (RECESS2 to RECESS5), as assessed by the professional forecasters chosen by the Philadelphia Fed.)²

3 Analysis of Yield Predictability

In this section we start by making a simple connection between the excess returns predicted by popular return-predicting factors and the prediction of future yields and future yield changes. By combining these predictions with the predictions afforded by the expectation-hypothesis, we show that overreaction in the formation of expectations offers itself as a natural explanation for our findings.

Let x_t^i be the i th of n sets of regressors. So, x_t^1 could be the regressors that make up the Cieslak-Povala cycles, x_t^2 could be five forward rates that make up the Cochrane-Piazzesi “tent”; etc. In what follows, we looked at the Cochrane-Piazzesi forward-rate regressors (Cochrane and Piazzesi, 2004), at the Cieslak-Povala cycles (Cieslak and Povala, 2010), and at the slope and level cycles described in Rebonato and Hatano (2018). We refer to each set of regressors as a ‘model’, and we label the models as CP , CiP , and RH for Cochrane-Piazzesi, Cieslak-Povala and Rebonato-Hatano, respectively.

Let $xret_t^{(N)}$ be the excess returns from time t to time $t + 1$ reaped from investing for one period at time t in an N -maturity bond:

$$xret_{t \rightarrow t+1}^{(N)} = p_{t+1}^{N-1} - p_t^N - y_t^1 \quad (1)$$

$$= (y_t^N - y_{t+1}^{N-1})(N-1) + (y_t^N - y_t^1), \quad (2)$$

where p_t^N denotes the time- t log-price of a discount bond of maturity N .

The regressions of these excess returns against the various return-predicting factors yield model- and maturity-dependent intercepts, $\alpha_i^{(N)}$, and slope coefficients, $\beta_i^{(N)}$,

$$xret_t^{(N)} = \alpha_i^{(N)} + \left(\beta_i^{(N)}\right) x_t^i + \epsilon_t^i \quad (3)$$

from which one can obtain

$$\mathbb{E}_t^i \left[xret_t^{(N)} \right] = \alpha_i^{(N)} + \left(\beta_i^{(N)}\right) x_t^i \quad (4)$$

²Note: The series RECESS2 is known as the “Anxious Index”.

where $\mathbb{E}_t^i [y_{t+1}^{N-1}]$ denotes the time- t expectation of the excess return obtained using regression model i , for $i = CP, CiP, RH$.

From Equation (2), the expected excess return from the carry strategy is also given by

$$\mathbb{E}_t [xret_t^{(N)}] = y_t^N N - \mathbb{E}_t [y_{t+1}^{N-1}] (N-1) - y_t^1 \quad (5)$$

and therefore the i -model time- t expectation of the time $t+1$ yield, $\mathbb{E}_t^i [y_{t+1}^{N-1}]$, is given by

$$\mathbb{E}_t^i [y_{t+1}^{N-1}] = \frac{y_t^N N - y_t^1 - [\alpha_i^{(N)} + (\beta_i^{(N)}) x_t^i]}{N-1}. \quad (6)$$

In particular, if the expectation hypothesis [EH] held true³, we would have

$$\mathbb{E}_t^i [y_{t+1}^{N-1}] = \frac{y_t^N N - y_t^1}{N-1} \quad [\text{EH}] \quad (7)$$

Given these expressions, after running the regressions in Equation (3) for each model we can calculate at time- t the $(N-1)$ -maturity yield at time $t+1$ as predicted by each model. We can compare these expectations with the realized future yield, y_{t+1}^{N-1} , and with the future yields predicted by the pure expectation hypothesis (as given by Equation (7)). The resulting time series are shown in Figs (1) and (2), for the 10-year and 3-year carry strategies (which therefore allow the prediction of the 9- and 2-year future yields).

More interesting than the predicted levels are the predicted changes in the 9- and 2-year yields, as predicted by the various models, and by the expectation hypothesis. We compare these as well with the realized changes. Fig (3) shows the term structure of correlation between the realized and expectation-hypothesis-predicted annual changes in yields of different maturities, and Tabs II to V show the correlations between the predicted yield levels and yield changes for the 2-year and 9-year yields.

9-year (levels)	RH	CP	CiP	EH	Real
RH	1.00				
CP	0.98	1.00			
CiP	1.00	0.98	1.00		
EH	0.96	0.98	0.96	1.00	
Real	0.96	0.94	0.96	0.92	1.00

Tab II: Correlation between the 9-year yield levels predicted by the Rebonato-Hatano (labelled ‘RH’), Cochrane-Piazzesi (‘CP’), Cieslak-Povala (‘CiP’) and pure expectation models (‘EH’) and realized yields (‘Real’).

³When it comes to the expectation hypothesis, there is general lack of agreement in the terminology between pure, local, weak and strong expectation hypothesis. We define expectation hypothesis in what follows the hypothesis that forward rates are unbiased predictors of one-period future yields, or, equivalently, that the risk premium should be zero (not a state-independent non-zero constant).

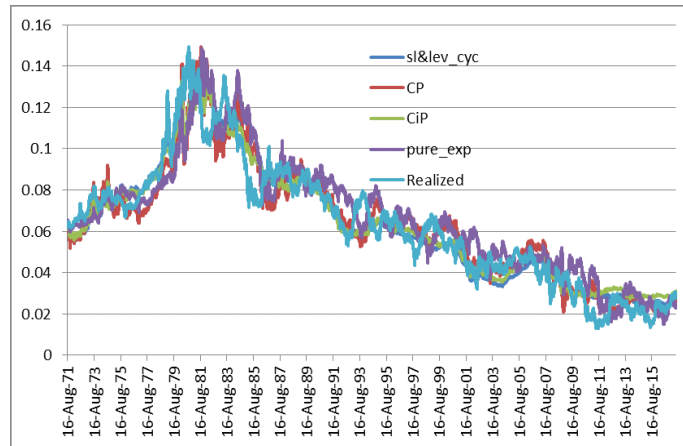


Figure 1: The expectations of the 9-year yield in 1 year's time produced by the CP, CiP and RH models (curves labelled CP, CiP and RH), and by the expectation hypothesis (curve labelled 'pure-exp'), and the realized 9-year yield in 1 year's time (curve labelled 'Realized').

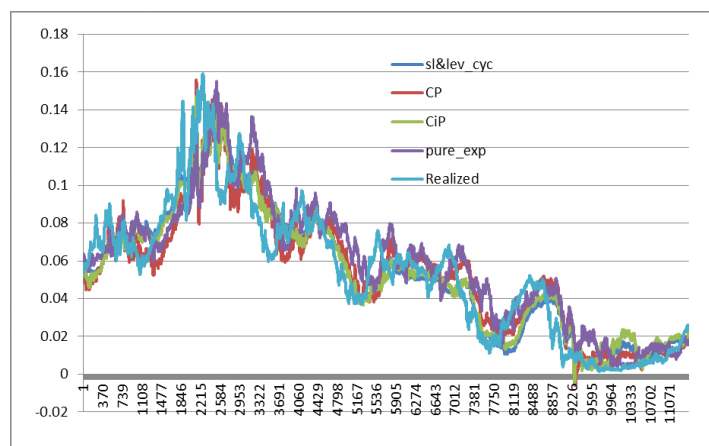


Figure 2: Same as Fig (1) for the 2-year yield.

9-year (changes)	RH	CP	CiP	EH	Real
RH	1.00				
CP	0.54	1.00			
CiP	0.94	0.69	1.00		
EH	-0.27	-0.43	-0.20	1.00	
Real	0.65	0.41	0.66	-0.18	1.00

Tab III: Same as Tab II for annual changes in yields.

2-year (levels)	RH	CiP	CiP	EH	Real
RH	1.00				
CP	0.97	1.00			
CiP	1.00	0.98	1.00		
EH	0.96	0.98	0.96	1.00	
Real	0.93	0.91	0.93	0.89	1.00

Tab IV: Same as Tab II for 2-year yield

2-year (changes)	RH	CiP	CiP	EH	Real
RH	1.00				
CP	0.72	1.00			
CiP	0.97	0.79	1.00		
EH	-0.64	-0.85	-0.66	1.00	
Real	0.30	0.04	0.34	0.02	1.00

Tab V: Same as Tab III for the 2-year yield changes.

Several observations are in order.

First of all we note the very strong similarity in predictions (both of levels and changes) between the Cieslak-Povala and the Rebonato-Hatano models. This is noteworthy given the very different ways to construct the two return-predicting factors. We discuss in the conclusions the relevance of this observation.

The next observation is that in terms of correlation between predictions and realizations of levels the expectation-hypothesis component of the level predictions (the term in Equation (7)) fares no better than the random-walk hypothesis. In contrast, all return-predicting factors predict levels better than the expectation hypothesis or the random-walk model. See Tabs II and IV.

The picture becomes more interesting when we look at the predictions of changes (see Tabs III and V). We see that the predictions of all the return-predicting factors are strongly correlated among themselves (again, with the CiP and the RH models showing particularly high correlation). The correlation between the model-predicted and the realized changes remains high for all the models (ranging from 0.41 to 0.66 for the 9-year yield). However, the correlation between the realized yield changes and the yield changes predicted by the expectation hypothesis is statistically significantly positive only for 1-year yield

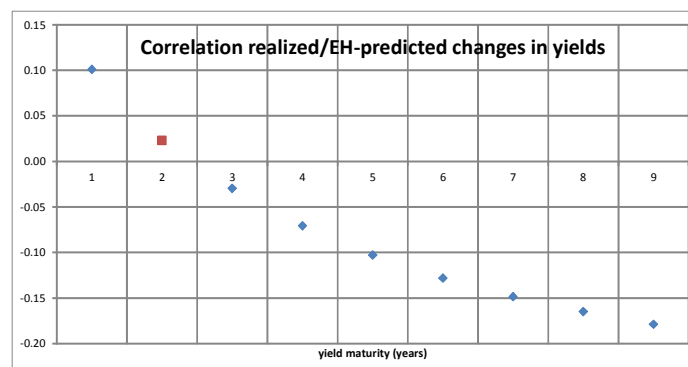


Figure 3: The term structure of correlation between the realized and expectation-hypothesis-predicted annual changes in yields of different maturities. The blue diamonds indicate the correlations that are different from zero at the 0.99 confidence level. The red square indicates the one correlation which is statistically indistinguishable from zero.

changes, and statistically significantly *negative* at the 99% confidence level for all maturities greater or equal to three years.

So, the bulk of the predictability of levels comes from where levels are today, with the return-predicting factors adding some additional predictability. But when it comes to yield changes, the negative correlation between the changes predicted from the expectation hypothesis and the realized changes shows that when the yield curve is upward sloping (and rates are therefore ‘EH-expected’ to rise) rates systematically increase by less than implied by the shape of the yield curve; and vice versa for a downward sloping yield curve. (The t statistic of the slope coefficient in the regression of the expectation-hypothesis-predicted average yield changes against the realized average yield changes is greater than 19, and the coefficient is a negative -0.032 .)

The predictions of the expectation hypothesis are therefore not just inefficient by virtue of missing a predictable component – for long maturities they are systematically biased in the sense of exaggerating by how much rates should increase or decrease. The effect becomes stronger, the longer the maturity.

This term structure of correlation between predictions and realizations of yield changes (shown in Fig (3)) should be compared with the explanatory power of excess returns afforded by the various return-predicting factors, as quantified by the R^2 of the regression (3). This R^2 is weakest and (for some models) barely significant, for short maturities, and larger and statistically significant for all models for long maturities. This prompts a natural question: is the degree of explanatory power of returns afforded by the different models linked to the degree of overreaction (as captured by the negative correlation between realized and expectation-hypothesis-predicted future yields)?

Indeed, a regression of the correlations between expectation-hypothesis pre-

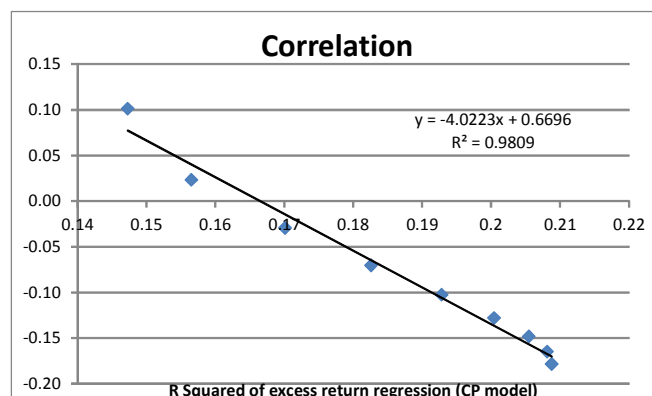


Figure 4: The correlations between expectation-hypothesis-predicted and realized yield changes (on the y axis) against the R^2 of the regression of excess returns against the CP return-predicting factors (five forward rates), on the x axis.

dicted and realized yield changes against the R^2 of the regression of excess returns against the CP return-predicting factors (five forward rates) has an R^2 of 0.98 (similar results are obtained for the other return-predicting factors): the maturities for which the CP factor predicts excess returns best are those with the most negative correlation between EH-predicted and realized yield changes. See Fig (4). Similar results were obtained for the other return-predicting factors.

We have shown that return-predicting factors predict returns and future yields better for those maturities for which the correlations between the realized and the expectation-hypothesis-predicted changes in yields are negative. If this negative correlation is indeed a result of overreaction, then the predicting power of return factors would stem from their ability to predict the reversion to the mean ('fundamentals') of an overreaction-driven expectation. Indeed, this interpretation finds corroboration in Cieslak and Povala (2010) who suggest that fleeting 'deviations from fundamentals' should be responsible for the predicting power of their cycles. Similarly, Rebonato and Hatano (2018) identify the source of extra predicting power of the new-generation factors (such as the CP, the CiP and the various cycles built in Rebonato and Hatano) with relatively high-frequency corrections to deviations from fundamentals, which are also compatible with overreaction. (See Section 4 for a more careful discussion of the deviations from rational Bayesian updating of expectations that may be at play).

Note also that the predictions of changes produced by all the return-predicting factors (which are strongly positively correlated with the realized changes) are strongly negatively correlated with the predictions of the expectation hypothesis. This also suggests that the predictability afforded by the various models could be well accounted for their ability to detect mean-reverting overreactions.

This possibility finds an indirect corroboration Tab VI, which shows the Sharpe Ratios for the unconditional carry strategy (bottom row), and for the conditional carry strategies whose notional is set proportional to the CP or CiP return-predicting factor.

Maturity	2	3	4	5	6	7	8	9	10
$SR(CiP)$	0.522	0.533	0.544	0.554	0.561	0.566	0.569	0.571	0.571
$SR(CP)$	0.408	0.427	0.441	0.450	0.457	0.461	0.464	0.466	0.467
$SR(unc)$	0.359	0.357	0.354	0.348	0.340	0.331	0.320	0.308	0.297

(8)

Tab VI: Sharpe Ratios of the conditional carry strategies with the notional proportional to the Cieslak-Povala (CiP) or Cochrane-Piazzesi (CP) return-predicting factors (first two rows), and of the unconditional carry strategy (third row) for maturities from 2 to 10 years.

As one can readily see, the unconditional strategy (last row) has Sharpe Ratios declining as a function of maturity. When the carry strategies are conditioned on the strength of the CiP or CP signals (first and second row, respectively), the trend in Sharpe ratios is reversed, with the greater predicting power of the return-predicting factors for longer maturities more than compensating for the decline in Sharpe ratio of the unconditional strategies. Admittedly, the increase in predicting power could be due to the fact that a risk premium (that is expected to increase approximately linearly with maturity) is more accurately detected for long than for short maturities. It can also be explained, however, by noting that the trend reversion implied by negative correlation between expectation-hypothesis-predicted and realized changes in yields observed for long maturities makes it easier for the return predicting factor to guess where yields ‘will go’. Indeed the improvement in Sharpe ratio with respect to the unconditional strategy in going from 2- to 10-year maturities goes from 25% to over 75% in the case of the CP signal, and from 60% to 115% in the case of the CiP signal.

At this stage, overreaction seems to provide a natural explanation of these preliminary findings, but the analysis so far does not exclude the possibility that risk premia may be at play as well – or, indeed, that the state dependence of risk premia may be such as to account for the observed negative correlations.

We therefore investigate in the next section whether it is possible to formalize the overreaction intuition in such a way as to obtain a more precise link between overreaction and the stylized empirical facts about the predictability of excess returns highlighted in the Introduction.

4 An Overreaction Model

In this part of the paper we want to explore whether a simple model with risk-neutral, overreacting investors can account for stylized facts about risk premia, as highlighted, for instance, in Tabs I and II.

We choose the model of overreaction ('diagnostic expectations') in Gennaioli and Shleifer (2018) because of its extremely parsimonious nature. This model of expectation formation builds on the representative heuristics of judgment formation under uncertainty (Kahnemann and Tversky, 1983), and has the features of being psychologically plausible, of being 'portable' in the sense proposed by Matthew Rabin (2013),⁴ of giving rise to forward-looking beliefs (ie, beliefs that are updated and changed as new information becomes available, although perhaps not in a statistically optimal manner), and of being testable against survey evidence of beliefs. The reader is referred to the discussion in Gennaioli and Shleifer (2018) for further details, but for our purposes the most important feature of representativeness is that investors tend to overweight the probability of future outcomes that become more likely as new information becomes available. In this model, investors react to macroeconomic innovations in the right direction, but too much. As alluded to above, a very desirable feature of this approach is that it can account for a number of cognitive deviation from rational Bayesian updating (such as, for instance, neglect of tail risk, base-rate neglect, conjunction fallacy, overreaction) in a very parsimonious manner.

Quantitatively, the model gives rise to the following predictions.

Let X_t be a financial quantity about which expectations are to be formed (say, the future value of the 10-year Treasury yield) in the light of background information, π_t , available up to time t (say, the history of inflation up to the time of expectation formation). Denote by $h(X = x|\pi_t)$ the true distribution of X_{t+1} , and define representativeness, $R(x, \pi)$, of realization of x given information π , as

$$R(x, \pi) = \frac{h(X = x|\pi_t)}{h(X = x|\tilde{\pi}_t)} \quad (9)$$

In the context at hand $\tilde{\pi}_t$ can be interpreted as information about the economy before the latest inflation information has become available, and therefore $\tilde{\pi}_t = \pi_{t-1}$.⁵ As the definition show, x is therefore more representative if it is relatively more frequent in $h(X = x|\pi_t)$ than in $h(X = x|\pi_{t-1})$. With representativeness, the most likely outcomes for inflation are those whose likelihood has increased the most on the basis of the extra information from π_t to π_{t-1} .

Let then X_t follow an AR(1) process

$$X_{t+1} = \rho X_t + \epsilon_{t+1} \quad (10)$$

with ϵ_{t+1} a zero-mean iid shock with constant variance σ^2 . The rational expectation formed at time t for the value of the process at time $t + 1$ is given by:

$$\mathbb{E}_t[X_{t+1}] = \rho X_t \quad (11)$$

In the absence of news, the state of inflation that would prevail at time t is ρX_{t-1} . The set $\tilde{\pi}_t = \pi_{t-1}$ is therefore the true distribution of X_{t+1} condi-

⁴Roughly speaking, portability means that the theory should account for the evidence from psychological experiments, from social judgements, from financial markets, and from all domains where individuals are called upon to make judgements and create expectations.

⁵See Gennaioli and Shleifer (2018) for a more general interpretation of $\tilde{\pi}_t$.

tional on the current state being ρX_{t-1} . Given the assumptions about ϵ_{t+1} , the distribution is normal with mean $\rho^2 X_{t-1}$ and variance σ^2 .

As Gennaioli and Shleifer (2018) show, representativeness then becomes

$$R(x, I) = \frac{h(X = x | \rho X_t)}{h(X = x | \rho X_{t-1})} . \quad (12)$$

We note for future discussion that the most representative future states (ie, the states for which the probability assessment has increased most) are in one of the two tails.⁶ Bordalo, Gennaioli and Shleifer (2018) then show that the representativeness-distorted density, $h^\theta(X = x | I_t)$, is given by

$$h^\theta(X = x | \pi_t) = h(X = x | \pi_t) \left[\frac{h(X = x | \rho X_t)}{h(X = x | \rho X_{t-1})} \right]^\theta Z \quad (13)$$

where $\theta \geq 0$ captures the extend of probability distortion by representativeness, and Z is constant that ensures normalization of the density. For $\theta = 0$ the ‘rational’ density is undistorted. For the case at hand, Gennaioli and Shleifer (2018) show that, under representativeness and under the assumptions above about the process X_t , the believed distribution for X_{t+1} is still normal, with the same variance σ^2 , but with a θ -distorted expectation given by

$$\mathbb{E}_t^\theta[X_{t+1}] = \rho X_t + \rho\theta(X_t - \rho X_{t-1}) . \quad (14)$$

Intuitively speaking, the representative density is distorted in the direction of current news, $X_t - \rho X_{t-1}$. If the process is positively autocorrelated, ($\rho > 0$), diagnostic expectations give rise to overreaction (extrapolation), in the sense that, after good news, the most representative states are those in the right tail, and vice versa.

This is the set up we need for the formalization of the predictability of excess return that we sketch below.

5 A Simple Monetary Model

In this section we present a stylized model of the monetary actions of a central bank and Treasury bond investors. We assume that the task of the central bank is to keep inflation close to a given target, and that, in order to do so, it monitors a quantity (which could be inflation itself, or GDP growth, or a combination thereof) and adjusts accordingly the Fed Funds as described in the following. We call the monitored macrofinancial quantity "inflation" (π_t), without being committed to a strict inflation interpretation. The monetary authorities monitor inflation and intervene on the Fed Funds every dt years. (For ease of derivation we start the treatment in continuous time, and then revert to discrete time.)

As for the Treasury bond investors, we assume that they are risk-neutral, and that they know the mandate and ‘model’ of the central bank. We have

⁶See Gennaioli and Shleifer (2018) for a proof, that extends to all distributions with monotone likelihood ratio properties.

a universe with ‘rational’ and a universe with ‘biased’ (affected by representativeness, overreacting) investors⁷. In both universes, the investors observe the monitored (inflation) macrofinancial quantity at the same time as the central bank.

In our model inflation is assumed to follow a process

$$d\pi_t = -Bdr_t + \sigma_\pi dz_t^\pi \quad (15)$$

where dz_t^π is the increment of a standard (zero-mean) Wiener process, as specified below. The term $-Bdr_t$ represents the change in inflation brought about by the actions of the monetary authorities. Since we are describing what will turn out to be a negative feedback process, it is important to keep track of the precise timing of the various events. So, π_t is the inflation observed at time t (no indexation lag), and $d\pi_t$ is the change in inflation from time t to time $t + \Delta t$:

$$d\pi_t = \pi_{t+1} - \pi_t \quad (16)$$

The monetary authorities have an inflation target rate, θ_π , and observe at time t whether inflation is above or below target, ie, they look at the deviation quantity, ξ_t :

$$\xi_t = \pi_t - \theta_\pi . \quad (17)$$

They move the Fed funds rate (the short rate in the following) by an amount proportional to ξ_t , and in the direction of hiking rates if the deviation is positive. More precisely, at time t , when the new reading of inflation π_t is revealed, the short rate, which had been set at time $t - \Delta t$, was at r_t , and is moved for the period $[t, t + \Delta t]$ to the new value r_{t+1} . So,

$$dr_t = r_{t+1} - r_t \quad (18)$$

and

$$dr_t = A\xi_t dt, \quad A > 0 \quad (19)$$

Note that the shock to the administered rate is proportional to the interval between interventions. This makes sense: if the Fed met weekly or yearly the size of their rate adjustment would presumably be different.

Combining Equations (15) and (19) we find that the shock to inflation imparted at time t by the rate move dr_t is given

$$d\pi_t = -AB\xi_t dt + \sigma_\pi dz_t^\pi \quad (20)$$

Therefore

$$\begin{aligned} d\pi_t &= -Bdr_t + \sigma_\pi \varepsilon \sqrt{dt} \\ &= -AB\xi_t dt + \sigma_\pi \varepsilon \sqrt{dt} \end{aligned} \quad (21)$$

⁷We do not consider a third possible universe in which the biased and rational investors interact in the same economy. This would lead us to explore the interesting but unrelated topic of limits to arbitrage.

and, since $d\xi_t = d\pi_t$,

$$\implies d\xi_t = - \underbrace{(AB)}_K \xi_t dt + \sigma_\pi dw_t \quad (22)$$

where dw_t is also a standard (zero-mean) Wiener process. This shows that, in continuous time, the deviation of inflation from the target rate, ξ_t , follows an Ornstein-Uhlenbeck process, with reversion speed K

$$K = AB$$

and zero reversion level. Discretizing the process for ξ_t as an AR(1) process, we have

$$\xi_{t+1} = \rho \xi_t + \epsilon_{t+1} \quad (23)$$

and

$$\mathbb{E}_t [\xi_{t+1}] = \rho \xi_t \quad (24)$$

with

$$\rho = e^{-K\Delta t} \quad (25)$$

where in discrete time we write x_{t+k} to signify $x_{t+k\Delta t}$.

5.1 The Unbiased Universe

The short rate that applies from time t to time $t+1$ is r_{t+1} . After observing the time- t reading of inflation, this is a known quantity at time t .

$$r_{t+1} = r_t + \Delta r_t = r_t + A\xi_t \Delta t \quad (26)$$

Then we have

$$r_{t+2} = r_{t+1} + \Delta r_{t+1} = r_t + A\Delta t (\xi_t + \xi_{t+1}) \quad (27)$$

and, in general,

$$r_{t+n} = r_t + A\Delta t \sum_{j=0, n-1} \xi_{t+j} \quad (28)$$

with $t_0 = t$.

In the universe where investors are unbiased, the expected value of the short rate at time $t+n$ is given by

$$\mathbb{E}_t [r_{t+n}] = r_t + A \sum_{j=0, n-1} \mathbb{E}_t [\xi_{t+j}] \Delta t \quad (29)$$

and, from Equation (24) we have

$$\mathbb{E}_t [r_{t+n}] = r_t + A\xi_t \sum_{j=0, n-1} \rho^{j\Delta t} \Delta t \quad (30)$$

Neglecting convexity, unbiased risk-neutral investors create at time t the price of a discount bond maturing at time $T = N\Delta t$, P_t^N , as

$$P_t^N \simeq \exp \left\{ - \sum_{k=1, N} \mathbb{E}_t [r_{t+k}] \Delta t \right\}$$

and therefore

$$\begin{aligned} P_t^N &\simeq \exp \left\{ - \sum_{k=1, N} \left[r_t + A\xi_t \sum_{j=0, k-1} \rho^{j\Delta t} \Delta t \right] \Delta t \right\} \\ &= \exp \left\{ -r_t T + \sum_{k=1, N} \left[A\xi_t \sum_{j=0, k-1} \rho^{j\Delta t} \Delta t \right] \Delta t \right\}. \end{aligned}$$

One can write for the yields

$$y_t^N = -\frac{1}{T} \log P_t^N \simeq r_t + \sum_{k=1, N} \left[A\xi_t \left(\frac{1}{N} \sum_{j=0, k-1} \rho^{j\Delta t} \right) \Delta t \right]. \quad (31)$$

Define

$$M^{(N)} \equiv \frac{1}{N} \sum_{j=0, k-1} \rho^{j\Delta t} \quad (32)$$

(This is a monotonically decreasing function of N , starting at 1 and asymptotically approaching 0.) Then one can write

$$y_t^N = r_t + \xi_t \sum_{k=1, N} A \times M^{(N)} \Delta t \quad (33)$$

These are the yields in the unbiased, risk-neutral universe.

5.2 The Universe with Representativeness-Biased Investors

The derivation of the yields in the case of biased investors proceeds along similar lines, but the formation of expectations is different, and, as shown above, one has

$$\mathbb{E}_t^\theta [\xi_{t+1}] = \rho \xi_t + \rho \theta (\xi_t - \rho \xi_{t-1}) \quad (34)$$

The same expression as Equation (29) for the expectation of the short rate n periods hence applies, but now with the iterated biased expectation in Equation (34).

$$\mathbb{E}_t^\theta [r_{t+n}] = r_t + A \sum_{j=0, n-1} \mathbb{E}_t^\theta [\xi_{t+j}] \Delta t. \quad (35)$$

It is possible to obtain concise expressions for the yields such as Equation (33), but these are rather unwieldy (and not terribly illuminating) and are therefore not reported. For the actual numerical calculations we directly used Equation (34) iteratively to obtain $\mathbb{E}_t^\theta [\xi_{t+j}]$.

5.3 Results

With the set-up described above we ran long-term simulations of the linked processes for the short rate and inflation, and calculated the yields in the biased and unbiased universes as derived above. For the simulations, we chose the following parameters:

- the elasticity of short rate moves to inflation shocks (ie, the quantity A in equation $dr_t = A\xi_t dt$) was taken to be equal to 2;
- the responsiveness of inflation to monetary tightening or loosening (ie, the quantity B in $d\pi_t = -Bdr_t + \sigma_\pi dz_t^\pi$) was taken to be equal to 0.5;
- the absolute volatility of inflation (ie, the quantity σ_π in $d\pi_t = -Bdr_t + \sigma_\pi dz_t^\pi$) was taken to be equal to 0.01.
- the inflation target (ie, the quantity in $\xi_t = \pi_t - \theta_\pi$) was taken to equal to 0.015;
- all simulations were started with initial values for the short rate and inflation of 0.025 and 0.02;
- three values for the representativeness coefficient in Equation (13) were considered ($\theta = 0, 0.25, 0.5$), with $\theta = 0$ corresponding to unbiased investors;
- all simulations were run for approximately 40 years, to roughly match the length of the data we use for the empirical studies reported in Sections 1 and 6.

As the round-number figures for the various parameters clearly indicate, no attempt was made to fine-tune the model. Rather, our intention was to explore whether a simple model such as the one outlined above could naturally recover salient features and stylized fact about the predictability of excess returns.

As a qualitative prelude to the discussion to follow, Fig (5) shows a representative path for inflation and the short rate, and Fig (6) the deviation of inflation from the target (blue curve) – ie, the quantity ξ in Equation (17)–, and the change in short rate applied as a result (red curve). One can observe the strongly mean reverting behaviour of inflation, and the positive autocorrelation of the ‘Fed funds’ process, two stylized features found in the real data.

Remaining at a qualitative level, Figure (7) then shows a snapshot of the yield curves obtained at a random point in time, in the unbiased case ($\theta = 0$) and with two degrees of biasedness ($\theta = 0.25$ and $\theta = 0.5$). For ease of comparability the same stream of random numbers was used to generate the yield curves in the two biased and the one unbiased worlds. The important thing to note is that the ‘biased’ curves are steeper than the unbiased curve. As we discuss at greater length below, overreaction has given rise to yields higher than they ‘should be’, and already at this stage it is plausible to expect that this will give rise a positive excess return for a carry strategy. We substantiate this expectation below.

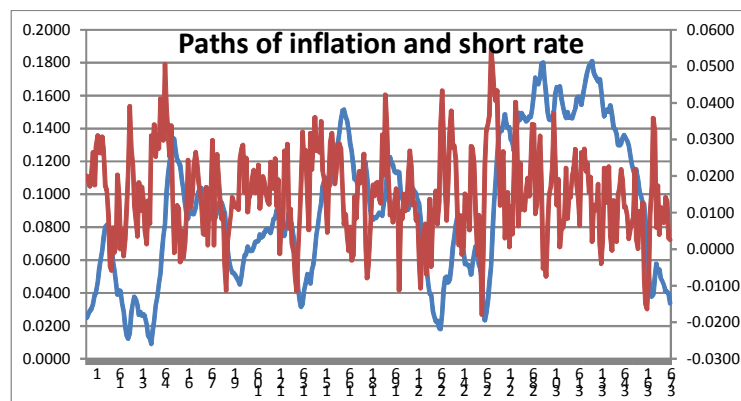


Figure 5: A simulation of the paths for inflation (right-hand vertical axis), and the short rate (left-hand vertical axis) obtained with the parameters specified in the text.

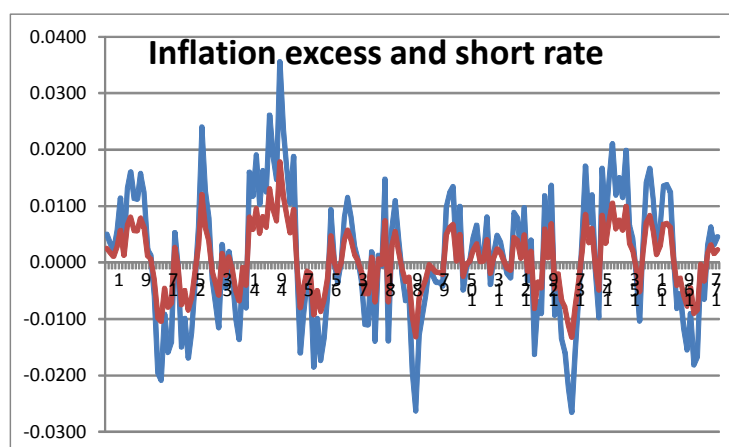


Figure 6: The deviation of inflation from the target (blue curve) – ie, the quantity ξ in Equation (17)–, and the change in short rate applied as a result (red curve)

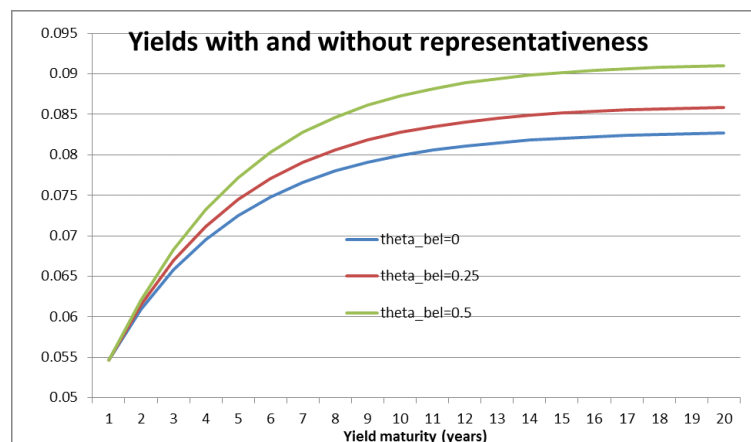


Figure 7: Three typical yield curves obtained assuming unbiased investors (curve labelled ‘theta_bel = 0’, corresponding to $\theta = 0$), and investors affected by increasing degrees of representativeness bias (overreaction), as reflected in values of $\theta = 0.25, 0.50$ (curves labelled ‘theta_bel=0.25’ and ‘theta_bel=0.5’).

First of all we test whether the dynamics of the Fed funds and of the yield curves produced by our feedback overreaction model recover some salient features of the empirically observed dynamics of relevance to our investigation. Of course, in assessing the realism of the model, one must keep in mind that no attempt was made to calibrate the model parameters, and that therefore only a qualitative or semi-quantitative agreement can be expected.

- First of all, the model produces a very strong autocorrelation in level for the Fed Funds rate as found in the empirical data (the empirical autocorrelation is 0.99, 0.97 and 0.95 for 1-, 2-, and 3-month lags). The model autocorrelation of the ‘short rate’ is even higher, but we must remember that we are dealing with a one-factor model.
- The empirical serial autocorrelation of changes in the Fed Funds is strongly positive for the first lag (35%), but turns negative for longer lags. The same qualitative behaviour is displayed by the changes in the short rate produced by the model (although the autocorrelation for the shortest lag is approximately twice as large as what observed in reality). The same switch from positive to negative autocorrelation in changes occurs on average for the model for higher lags than with real data, but in more than 50% of the simulations we observe the same oscillatory pattern with sign switches in autocorrelation found with real data.
- The model produces a much stronger mean reversion for the slope than for the level of the yield curve. Indeed, depending on the simulations, the level can be a weakly mean-reverting or a borderline unit root process, while the

slope has reversion speed of the order of unity (half-life of approximately half a year). This also compares well with statistical estimations (see, eg, Diebold and Rudebusch, 2013).

These features suggest that our model of monetary actions is at least plausible. As we shall show, they are relevant to an overreaction explanation of the excess return predictability, to which we turn next.

We simulate with the parameters reported above many realizations of the model, and for each simulation we calculate the excess returns from the carry strategy for maturities from 2 to 5 years in quarterly maturity intervals. We note for future discussion that it is reasonable to associate in our model periods of high inflation in the model with expansions, and periods of low inflation as recessions. This will give us a way to test the business-cycle dependence of excess returns predicted by our model.

When the investors are rational Bayesian updaters of information ($\theta = 0$), the unconditional excess returns from the carry strategy are for 90% of the simulations statistically not significantly different from zero. Conditional returns (conditioned on the realization of the slope) are statistically significantly different from zero for only 5-to-10% of the simulations, but the R^2 of the regression of excess returns against the slope is extremely low (of the order of 0.02). So, unsurprisingly, neither unconditional nor conditional carry strategies would be profitable if risk-neutral investors did not overreact. Needless to say, this is an obvious result, and simply creates a benchmark of statistical accuracy for the simulations.

The situation changes dramatically when investors are affected by representativeness ($\theta = 0.25$, $\theta = 0.5$). Now

- the conditional carry strategy becomes profitable for more than 90% of the simulations;
- the R^2 of the regression of excess returns against the slope ranges from 0.08 to 0.20 (values that compare very favourably with the empirical values – see the x axis in Fig (4));
- the coefficients of the regression of the excess returns against the slope are significant (intercept and slope) for all maturities;
- the slope has greater explanatory power than the level. The ratio of the R^2 for the slope over R^2 of the level is somewhat higher in actual data than with the simulated data, but we must remember again that the model under study is a one-factor model;
- if, as mentioned above, we give a business-cycle interpretation to the realized inflation (which in our model displays positive serial autocorrelation) the business-cycle dependence of the excess return is recovered (see below);
- most importantly, the slope coefficient in the excess return regression against the yield curve slope are positive and strongly significant for all

maturities, indicating that a positively sloping yield curve is a harbinger of excess returns for the carry trade (and vice versa if the yield curve is downward sloping). This has been the most robust and universally accepted result on the predictability of excess returns, all the way back from the early studies by Fama and Bliss (1987) and Campbell and Shiller (1991).

Looking back at Fig (7) in the light of these findings, it is clear that overreaction gives rise to predictability by making yield curves higher than ‘they should be’ if upward sloping, and vice versa if downward sloping. So, for instance, if we interpret literally the stylized inflation story associated with our model, investors overestimate future deviations of inflation from the target rate in both directions. With a negative inflation shock (the onset of a recession) overreacting investors correctly expect the Central Bank to cut rates (the ‘kernel of truth’ in the Gennaioli and Shleifer (2018) account), but overestimate the extent of the easing, and predict future short yields lower than they should predict if not affected by representativeness bias. This creates an ‘excessively’ steep yield curve, and expected profits for the carry strategy. As this occurs when rates are cut (broadly speaking, at the onset of the recession) and the opposite would happen with a positive inflation shock (the last phase of an expansion), this feature recovers the business-cycle dependence of unconditional excess returns clearly shown in Tab II.

The results presented in this section suggest that it is possible to account at a semi-quantitative level for many of the stylized facts about the predictability of excess returns highlighted in Section 1 by invoking an overreaction model. This does not settle the question, however, of whether the overreaction or the risk premium explanation is more convincing. We try to answer this question by looking in the next section at whether the timing of the excess returns from the carry strategy are consistent with a risk-premium explanation of their origin.

6 Testing the Risk-Premium Explanation

Recall from the Introduction (see Tab I) that the sign of the returns from the carry strategy changes with the business cycle. If this predictability is to be explained in terms of a rational, unbiased investor demanding a risk premium, this risk premium must also change in sign. As long as investors do not become risk-seeking, this change in sign can only come from a change in correlation between the returns from the carry strategy and consumption. This is what we test in this section. The problem can be most simply formalized in a two-period economy with separable utility function as follows.

If we denote by S_t the price of a security at time t , by x_{t+1} its payoff at time $t + 1$, by R_f the gross riskless return, and by m_{t+1} the stochastic discount factor, one can write:

$$S_t = \frac{1}{R_f} E[x_{t+1}] + \underbrace{\text{cov}[m_{t+1}, x_{t+1}]}_{\text{risk premium}} \quad (36)$$

Dividing through by S_t , rearranging terms and using the definition of stochastic discount factor, m_{t+1} , one obtains

$$\frac{1}{R_f} E[R_{t+1}] = 1 - \text{cov}[m_{t+1}, R_{t+1}] = 1 - \text{cov}\left[\beta \frac{u'(C_{t+1})}{u'(C_t)}, R_{t+1}\right] \quad (37)$$

with

$$\beta = \exp(-\delta), \quad (38)$$

δ denoting impatience and $u(C_t)$ a time-separable utility function of time- t consumption. If, for simplicity, we use a utility function of the logarithmic type, $u(C) = \log(C)$, one gets

$$\frac{1}{R_f} E[R_{t+1}] = 1 + \text{cov}\left[\beta \frac{C_{t+1}}{C_t}, R_{t+1}\right] \quad (39)$$

and therefore

$$E[xR_{t+1}] = \text{cov}\left[\frac{C_{t+1}}{C_t}, R_{t+1}\right] \frac{1+r_f}{1+\delta} \quad (40)$$

which shows that the sign of the expected excess return, $E[xR_{t+1}]$, from a generic security depends on the sign of the covariance between the gross return from the security itself and the consumption growth.

If we made the very crude assumption that all consumption comes from the return from the market portfolio, and that the market portfolio is well approximated by a diversified equity index (such as the S&P 500 index), we would have

$$E[R_{t+1}] = R_f + \frac{1+r_f}{1+\delta} \text{cov}[R_{S\&P}, R_{t+1}] = R_f + \frac{1+r_f}{1+\delta} \sigma_{S\&P} \sigma \text{corr}[R_{S\&P}, R_{t+1}] \quad (41)$$

where $\sigma_{S\&P}$ and σ signify the volatilities of the S&P500 index and of the asset under consideration, respectively, and $R_{S\&P}$ is the return from the S&P 500 index, and $1+r_f = R_f$.

In what follows we attempt to create a more satisfactory consumption proxy, but the intuition does not change, and Equation (41) gives a convenient lens through which we can look empirically at excess returns: if positive excess returns are indeed due to our aversion to receiving good payoffs when we don't need them, and poor payoffs when we feel poor, then the correlation between the returns from the market and the returns from the asset in question should be positive. If, instead, an asset acts as a portfolio diversifier, then the correlation will be negative, and the asset should yield *less than* the risk rate. Furthermore, it is important to stress that, barring the possibility of investors becoming risk-seeking rather than risk-averse, a change in sign in the risk premium *can only come from a change in the correlation*. This simple observation is key to our discussion. In the empirical analysis that follows, Equation (41) will be generalized to

$$E[R_{t+1}] = R_f + \frac{1+r_f}{1+\delta} \text{cov}[R_{\text{Cons Proxy}}, R_{t+1}] = R_f + \frac{1+r_f}{1+\delta} \sigma_{\text{Cons Proxy}} \sigma \text{corr}_t[R_{t+1}^{\text{Cons Proxy}}, R_{t+1}] \quad (42)$$

where the subscript *Cons Proxy* denotes the chosen consumption proxy. We describe the construction of the consumption proxy in the next section.

7 Data and Methodology To Test the Risk-Premium Explanation

In what follows we show the results for the correlation between the rolling correlation, $corr_t \left[R_{t+1}^{Cons Proxy}, R_{t+1} \right]$, and the excess returns. When it comes to the correlation $corr_t \left[R_{t+1}^{Cons Proxy}, R_{t+1} \right]$ we stress again that we are interested in the expected correlation at the time of price formation, not in how well the expected correlation predicts the realized correlation: if investors expect at time t an asset to act, say, as diversifier, they will bid up its time- t price and depress the $[t, t + 1]$ return, irrespective of whether the expected correlation will materialize or not. We therefore estimate the correlation using information only available up to time t , and we make the assumption that investors observe the variability of the consumption-Treasury correlation, and extrapolate into the immediate future a medium-to-short-term rolling correlation estimated using the last three years worth of data.⁸ For the numerical estimate of the time- t correlation between the Treasury excess returns and the growth in the consumption proxy, $corr_t \left[R_{t+1}^{Cons Proxy}, R_{t+1} \right]$, we simply used a rolling fixed-length window of monthly data:

$$corr_t \left[R_{t+1}^{Cons Proxy}, R_{t+1} \right] = \sum \frac{\frac{1}{n} \left(R_{t-k}^{Cons Proxy} - \bar{R}^{Cons Proxy} \right) (R_{t-k} - \bar{R})}{\sqrt{var^{(n)} (R^{Cons Proxy}) var^{(n)} (R)}} \quad (43)$$

where n is the number of months in the estimate (see below), $R_{t-k}^{Cons Proxy}$ denotes the growth in consumption observed k months before the estimation month, t , R_{t-k} is the Treasury return also observed k months before the estimation month, t , $\bar{R}$ and $\bar{R}^{Cons Proxy}$ are the average Treasury returns and consumption growth over the n months before time t , and $var^{(n)} (R^{Cons Proxy})$ and $var^{(n)} (R)$ are the variances of the same quantities over the same n -month period.

Our ultimate goal is to obtain at each point in time an estimate of expected correlation over the next investment period. We make the assumption that investors take the past correlation calculated using Equation (43) as their prediction of the correlation over the next investment period. We would therefore like our estimate to be responsive (after all, we want to capture changes in the sign of the correlation), but not ‘hyperactive’. To achieve a reasonable compromise between responsiveness and stability of the correlation estimate we used for the length of the window a number of months, n , ranging from 36 to 90.

⁸The choice of three years is not crucial for the results. Similar choice for different data windows gave qualitatively comparable results.

After the ‘burn-in’ period, under our assumptions, this procedure gives us at each sampled point in time the expected correlation between the Treasury returns and consumption growth. We then regressed this quantity against the maturity-dependent excess returns.

Since excess returns are very noisy, we also performed the same calculation using, instead of Treasury excess returns, selected return-predicting factors. The reason for this choice is that, if these were correctly specified, they would identify the systematic component of Treasury excess returns without the large and unavoidable noise, and therefore provide a better estimate of the correlation between the (sign of the) risk premium and the correlation expectation estimated using Equation (43).

There are, of course, far more sophisticated and econometrically satisfactory ways to estimate correlation, but we have used the simplest approach to reduce the argument to its bare bones. We checked that using more sophisticated estimation procedures (such as assigning exponentially decaying weights to the observations) did not change the qualitative results – and, after all, since we are interested in the investors’ predictions of correlation over an observation period starting in the 1970s, there is no point in using estimation techniques (such as Engle’s anticipating correlations) that had not yet been invented at the time of price formation!

As for the return-predicting factors, we looked at three of the most popular ones, or with the most illustrious pedigree: the traditional yield-curve slope (which we proxy as the difference between the 10- and the 2-year yield); the Cochrane-Piazzesi (2005) forward-rate-based factor; and the Cieslak-Povala (2010) inflation-cycle-based factor. Given the similarity between the Cieslak-Povala and the Rebonato-Hatano return-predicting factors highlighted in Section 2, we drop in this section the Rebonato-Hatano factor, and add instead the simple slope factor (similar in construction to the slope factor in the early papers by Fama and Bliss (1987) and Campbell and Shiller (1991)).

7.1 Construction of the Consumption Proxy

It is well known that constructing a good proxy for consumption is a difficult task. We explore different ways to do so, and we discuss in what follows the robustness of our results to the different choices. As we shall see, the results show moderate dependence on the exact choice of consumption proxy.

One often quoted proxy for consumption growth is the absolute return from a broad and diversified equity index, such as the S&P500. A justification for this choice is easy to find if we assume that consumption fully comes from dividends and capital appreciation from an investor’s equity portfolio. (Indeed, this is one of the many ways in which the CAPM model is derived.) One appealing feature of this proxy is that there is virtually no measurement error. The main drawback is that S&P returns are very volatile, and include changes not just in expected future cashflows, but also in risk premia.

To the extent that equity prices may be too volatile to proxy effectively for expected consumption, we also considered as a possible proxy the current

GDP growth. This choice is justified by the observation that GDP growth acts as a reasonable proxy for real personal consumption expenditure (see, eg, Shefrin, 2008⁹). Given the strong serial correlation of GDP growth, we have taken either the current level of GDP growth or its two-year moving average as a reasonable estimate of GDP forecast. Diacon and Maha (2015) investigate links between consumption and GDP per capita, and find the two quantities to be co-integrated.

Real disposable income has also been often proposed as a proxy for consumption. We have therefore also considered yearly changes in real personal income per capita (US, chained 2012 dollars, seasonally adjusted) as a possible proxy for consumption.

Finally, we used the so-called ‘anxious index’. The Survey of Professional Forecasters asks panelists to estimate the probability that real GDP will decline in the current quarter, and in the four following quarter. There are therefore five such assessments, and the second (relating to the probability of a recession in the next quarter) is referred to as the ‘anxious index’. The anxious index is the probability of a decline in real GDP in the quarter after a survey is taken, and, for our purposes, it therefore has desirable forward-looking properties. Since we are looking at consumption growth, we have taken one minus the anxious index as one of our proxies of future consumption growth.

In our analysis we have both used the four proxies in isolation (call them $proxy_t^i$, $i = 1, 2, 3, 4$), and various convex combinations of the proxies, of the form

$$ConsIndex_t = \omega^T proxy_t^i \quad (44)$$

for several positive weights ω adding up to one. We found that the correlations $corr[R_{Cons Proxy}, R_{t+1}]$ that we are ultimately interested in did not behave qualitatively differently when we used the convex combination (44) or when we interpolated from the results obtained for the individual proxies. Therefore in the following section we will simply report the results for the four ‘corner’ cases, ie, for the four cases where the vector ω is made up of three zeros and one single ‘1’, varying in position from the first to the fourth.

Finally, some of the time series considered are highly volatile, and known with measurement error. When one wants to estimate the current trend of a time series, but is uncertain about the true data-generating process, and wants to make sure that the estimator is robust across different models, Evans, Honkapohja and Williams (2010) and Evans and Honkapohja (2009) show that the estimate given by

$$TRproxy_t^k = \frac{\sum_{i=0}^{t-1} v^i proxy_t^k}{\sum_{i=0}^{t-1} v^i}. \quad (45)$$

is optimal and maximally robust. (In Equation (45) the weight parameter, v , is

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set to 0.987, and the sum runs over 120 monthly observations.)¹⁰ We therefore also create a trend estimate for the S&P, GDP Growth and Income Growth proxies using Equation 45. (It does not make sense to create such a trend estimate for the professional forecasters' survey.)

None of these proxies are fully satisfactory, and the construction of a reliable consumption index is, of course, a major task in itself, and novel ingenious suggestions as to how to proxy consumption are continually being proposed. For instance, promising results towards the explanation of the equity premium puzzle have recently been obtained by using garbage or satellite data on lights at night (Henderson, Storeygard and Weil, 2012) as proxies for consumption.¹¹ These novel proxies have been shown to be particularly useful to gauge the GDP for countries for which the reliability of income accounts and domestic price indices is poor (the 'C' and 'D' countries in the Penn World Tables). In our study, however, we focus on US data, and, for all its well-known imperfections, the US GDP measure is among the most reliable (definitely an A country in the classification above).¹²

In any case, in this study we are simply interested in a qualitative explanation of the excess returns as a risk premium, and in an understanding whether a simple proxy can account for the regularities observed. We hope that, by using a variety of proxies, we can at least reach a qualitative understanding of whether a risk-premium explanation of the observed predictability of excess returns is plausible and convincing. After all, if finding a reasonable proxy were well nigh impossible, then the falsification of Equation (41) would become impossible.

7.2 Results for the Test of the Risk-Premium Explanation

Before entering a detailed discussion of the results, a high-level analysis already clearly shows how difficult it is to account for excess returns by invoking risk premium as an explanation. For all the proxies considered above, the average of the long-run running correlation between average Treasury returns and each consumption proxy turned out to be negative. For one proxy (GDP growth) the running correlation is negative on every single month covered in our data set. For the other proxies the running correlation is negative on 70% (S&P), 78% (Income Growth) and 73% (Inverse Anxious index) of the months. On the basis of this, Treasury appears to act as powerful insurance against changes in consumption. However, over the same period of almost half a century the empirical unconditional returns are positive at all maturities. The sign of the unconditional returns is therefore the opposite of what asset-pricing theory suggests it should be. This is (yet another) consumption puzzle – a puzzle that

¹⁰We note in passing that this is the estimator of inflation trend used by Cieslak and Povala (2010) in their study of excess returns in Treasuries.

¹¹Unfortunately these proxies do not extend to the early 1970s.

¹²The Penn World Tables are a standard compilation of across-country data on income. In the tables countries are given subjective grades on data quality, with grades A, B, C and D corresponding to an estimated margin of error of 10%, 20%, 30% and 40% or more, respectively.

is not resolved, as we show in the rest of this section, by carrying out a more detailed conditional analysis.

The more precise analysis is carried out as follows. We regressed the excess return from the carry strategy (averaged over maturities), avg_xret_t , against the running correlation, $avg_rho_t^{\{w\}}$, between the Treasury returns (also averaged over maturities) and the chosen consumption proxy:

$$avg_xret_t = \alpha^0 + \beta^0 \times avg_rho_t^{\{w\}} + \varepsilon_t^0 \quad (46)$$

with $\{w\}$ the weights used in the construction of the proxy. We also regressed the excess return from the carry strategy (averaged over maturities) predicted by the three models against the running correlation between the three chosen return predicting factors (also averaged over maturities) and the chosen consumption proxy:

$$avg_xret_t^k = \alpha^k + \beta^k \times avg_rho_t^{\{w\}} + \varepsilon_t^k \quad k = 1, 2, 3 \quad (47)$$

with $k = 1$ corresponding to the CiP factor, $k = 2$ to the slope factor, and $k = 3$ to the CP factor.

The results are shown in Tabs VIIa to VIId, for the four ‘corner cases’, ie, for the cases where the weights were set to one for one proxy, and zero for all the other proxies. Interior points did not produce unexpected results, and are therefore not reported. A negative (positive) beta indicates a sign for the slope inconsistent (consistent) that the risk-premium explanation of the excess returns.

Treasury returns	β^0	R_0^2
<i>GDP</i>	−0.125***	0.104
<i>S&P</i>	0.029	0.002
<i>Income</i>	0.020	0.002
<i>1 − Anxious</i>	0.062***	0.024

Tab VIIa: The β^0 coefficient in the regression (46), and the associated R^2 . Three, two or one stars indicate results significant at the 99.9, 99 or 95% confidence level.

Cip RPF	β^1	R_1^2
<i>GDP</i>	0.053***	0.025
<i>S&P</i>	−0.052***	0.070
<i>Income</i>	−0.035***	0.019
<i>1 − Anxious</i>	−0.009	0.002

Tab VIIb: The β^1 coefficient in the regression (46), and the associated R^2 . Three, two or one stars indicate results significant at the 99.9, 99 or 95% confidence level.

Slope	β^2	R_2^2
<i>GDP</i>	-0.041***	0.051
<i>S&P</i>	0.026***	0.014
<i>Income</i>	-0.027***	0.040
$1 - \textit{Anxious}$	0.004	0.001

Tab VIIc: The β^2 coefficient in the regression (46), and the associated R^2 . Three, two or one stars indicate results significant at the 99.9, 99 or 95% confidence level.

CP RPF	β^3	R_3^2
<i>GDP</i>	-0.005	0.001
<i>S&P</i>	-0.032***	0.019
<i>Income</i>	-0.033***	0.036
$1 - \textit{Anxious}$	0.014*	0.010

Tab VIId: The β^3 coefficient in the regression (46), and the associated R^2 . Three, two or one stars indicate results significant at the 99.9, 99 or 95% confidence level.

As the tables show, all the β coefficients with the highest R^2 (0.104, 0.070, 0.0512 and 0.0355) are negative. Seven out of ten of the β coefficients statistically significant at the 99.9 confidence level are negative, indicating again the opposite relationship between excess returns or return-predicting factors to what a risk-premium explanation would suggest. And, overall, the power of the consumption/return correlation to explain excess returns, as quantified by the R^2 , turned out to be always very low.

These findings pose an obvious problem: As Tabs VIIa-d show, if the excess returns were due to a risk premium, then the sign of this risk premium would have to be state-dependent – indeed, Tab II shows that the sign and magnitude of the excess returns display a clear dependence on the state of the economy. A market price of duration risk with varying sign should be positive or negative (with concordant sign) depending on the sign of the correlation on the right-hand side of Equation (41). The estimated correlations do not support this explanation at all – if anything, whatever is statistically significant strongly tends to have the wrong sign.

Consumption-based asset pricing studies have notoriously been fraught with difficulties: see, for instance, the equity premium puzzle, the short rate puzzle and the excessive stock volatility puzzle and Campbell (2018) for a discussion.) However, in all these problematic applications of a consumption-based asset-pricing approach, while the quantitative match leaves a lot to be desired, at least the qualitative behaviour is recovered: the theoretical equity risk premium, for instance, is too small, but at least it has the right sign; the same applies to the theoretical prediction of the short rate. In the application at hand we find either a lack of explanatory power, or even the opposite sign.

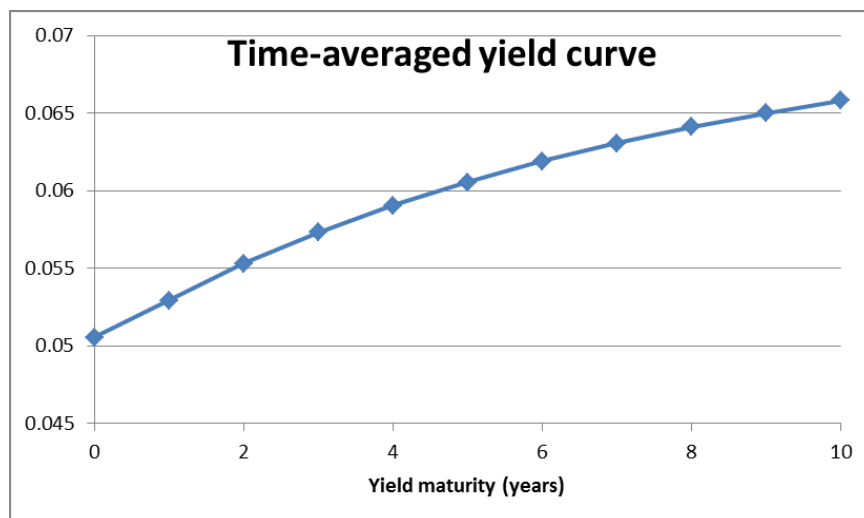


Figure 8: The time-averaged yield curve obtained with the data described in Section 00. Each point is the time average of the yield for the maturity on the x axis.

Proponents of a rational-investor, asset-pricing-based explanation of the predictability of Treasury returns often point to the upward-sloping shape of the time-averaged term structure of yields as a convincing indication that risk premia should be at play. We discuss this objection in detail in the next section.

8 Why Is the Average Yield Curve Upward Sloping?

If one averages over time the yields that have occurred from 1971 to 2018 one finds an upward sloping average term structure, in which the difference between the 10- and the 1-year yield is almost 130 basis points. See Fig (8). How likely is it that, in the absence of a positive risk premium, a forty-year-averaged should display such a steep slope?

If one assumes that the slope follows a Ornstein-Uhlenbeck process *with serially independent increments*, dz_t ,

$$dsl_t = \kappa_{sl} (\theta_{sl} - sl_t) dt + \sigma_{sl} dz_t \quad (48)$$

one can simulate monthly realizations of the slope out to a 50-year horizon assuming a reversion level of zero, calculate the average slope of the yield curve, and repeat the procedure many times. (As elsewhere in this paper, for this test we define the slope as the difference between the 10- and the 1-year yields.)

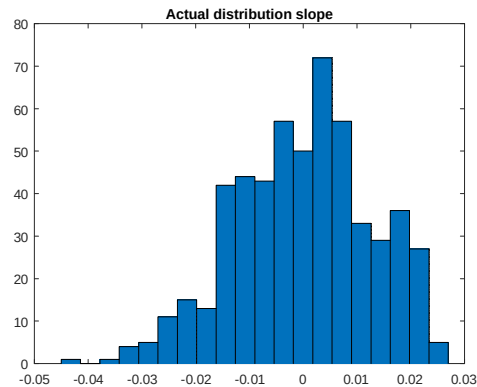


Figure 9: Actual distribution of the yield curve slope for the data described in Section 00 (period 1071:2017).

In order to carry out this simulation we estimated for the Ornstein-Uhlenbeck properties of the slope process a reversion speed of 0.45 years^{-1} ; we simulated 10,000 times a 50-year discretization of the process; and we calculated the average slope at the end of each of the 10,000 50-year simulations.

The results are shown in Figs (9) and (10) for the historical and simulated distributions of yield-curve slope, respectively. We then constructed an empirical cumulative distribution for the realizations of the average slope. The complementary empirical cumulative distribution thus obtained gives the total probability of obtaining by chance a slope greater than any value – in particular, greater than 128 basis points. Irrespective of the details of the simulation (for instance, we allowed for the volatility of the slope to be dependent on the slope level, and for a GARCH(1,1) volatility for the slope), we never obtained a cumulative probability for a slope greater than 128 basis points higher than 0.5%. Taking convexity into account would, if anything, make the result stronger.

Since the shape of the yield curve can be thought of as the sum of the expectation, risk premium and convexity contributions, and since expectations over 50 years should average out to zero, at first brush this seems to provide compelling evidence that a substantial positive risk premium must be at play. Indeed, the slope of the average yield curve had been frequently adduced as indirect evidence that risk a positive premium must contribute significantly to the slope of the yield curve. And since rates have substantially and unremittingly fallen since the 1980s, the argument goes, it is implausible to assume that expectation should have been pointing *upwards*.

The results however, strongly depend on the assumptions that the increments of the Ornstein-Uhlenbeck process in Equation (48) should be independent. But the slope is calculated as the difference between the 10-year yields (whose increments show little serial autocorrelation) and the 1-year yield, which inherits

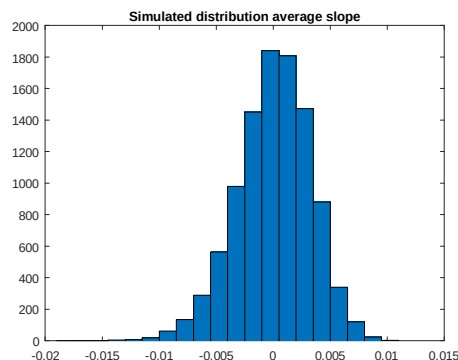


Figure 10: Simulated distribution of the yield curve slope assuming an OU process (see Equation (48)) for the slope *with independent increments*. Under the assumption of lack of serial correlation of the increments, the parameters for the simulation were obtained using the same data with which Fig (9) was obtained.

a substantial amount of serial auto-correlation from the Fed Fund rates. See the discussion in Section 5.3. So, by construction, the slope also displays a significant serial autocorrelation. (Regressing changes in slope with lags from 1 to 12 months, shows correlation coefficient for lags of 1, 2, 4, 7, 8, 11 and 12 months statistically significant at the 99% confidence level.)

In reality, as Fig (11) clearly shows, since 1971 (the start date for our data), we have only had approximately half a dozen major tightening and easing cycles. Now, during an easing cycle, which occurs in situations of economic distress or incipient disinflation, short-dated yields (which have a higher volatility than long-dates yields, and are more strongly linked to the rates affected by the central bank) fall more than long-dated yields. This automatically generates an increase in the slope of the yield curve, *even if the overall level of rates is in general falling*. The converse happens in a tightening cycle, which produces an upward sloping yield curve.

Since we only have had half a dozen major tightening and easing cycles, we cannot hope the law of large numbers to work out its magic, and out of as few as 12 cycles one can easily have a significant imbalance between tightening and loosening cycles. Indeed, we tested this imbalance by counting how many cuts or hikes in the Fed Funds greater than a certain threshold we have had since 1971. We set the threshold as a multiple of the standard deviation of the changes in the Fed funds time series.

For any reasonable threshold, we find that the number of rate cuts has exceeded the number of rate hikes. This is shown in Fig (12) for a variety of values of the threshold. (Since threshold was chosen to be the number of standard deviation shown on the x axis, a value of the threshold of, say, 1.5

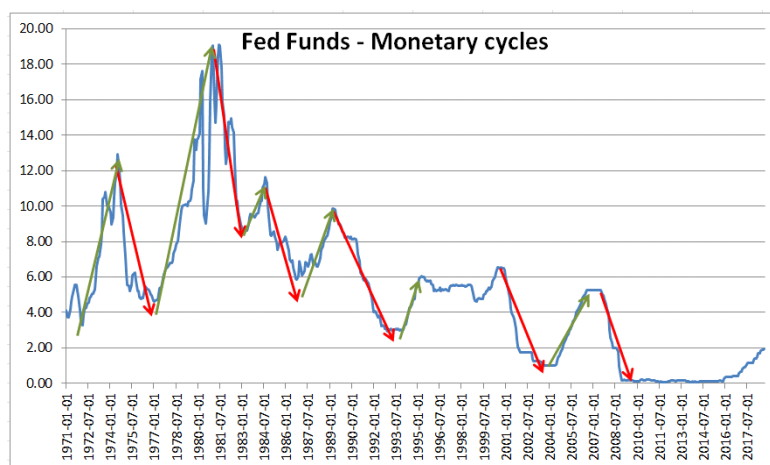


Figure 11: The Fed Funds rates from 1971 to the end of 2017, with the major tightening cycles shown by the green upward pointing arrows, and the easing cycles by the red downward pointing arrows. These arrows have been inserted to guide the eye, but subjective the classification of the cycles has not been used in the quantitative results discussed in the text. (One could insert, for instance, one extra tightening and one easing cycles around 1980, but we have chosen not to show either for clarity of presentation.)

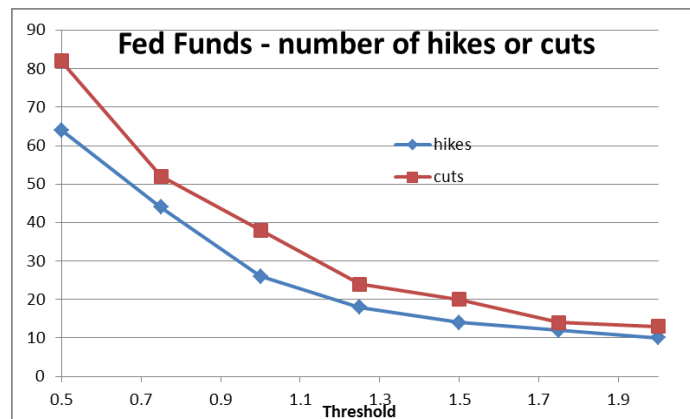


Figure 12: The number of rate hikes (blue diamonds) or cuts (red squares) as a function of the threshold described in the text. Note that there have been more cuts than hikes at all levels of the threshold.

means that we only counted hikes or hikes greater than 1.5 times the standard deviation of the Fed Funds changes.) We also find that, for all levels of the threshold between 0.5 and 2, the average size of rate cuts is greater than the average size of rate hikes.

From this analysis the picture therefore emerges of a predominance of easing over tightening cycles over the short half century under study, and of more aggressive cuts than hikes. As easing monetary operations produce a steepening of the yield curve, and tightening operations an inversion, we can confidently attribute the empirically observed positive slope of the average yield curve to the fact that, on average, we have had more, and more severe, easing than hiking cycles.

Finally, recall that the usual explanation for the dependence of the magnitude of the (positive) risk premium on the slope invokes heightened risk aversion (and so a high risk premium) at the onset of market distress, which prompts the monetary authorities to cut rates. Conversely, the zero or negative risk premium with upward-sloping yield curves is explained in terms of reduced risk premium (high 'animal spirits') in periods of expansions. In reality, the biggest losses incurred by a carry trade occurred in the late 1970s, when the Fed under the guide of then-Chairman Volker tightened rates to unprecedented levels. These aggressive tightening was prompted by the desire to keep under control a stubbornly rising inflation. Now, for the holders of fixed-income securities, few periods are more worrisome than times of difficult-to-control inflation. It is difficult to believe that exactly when it seemed most difficult to control inflation the risk aversion of investors should have become zero or negative. Yet, if one excludes the late 1970s hiking period, the evidence for the link between the magnitude of the risk premium and the slope of the yield curve becomes much

weaker.

9 Conclusions

In this paper we have examined whether a risk-premium explanation or an explanation based on a specific behavioural cognitive bias can better account for the predictability of excess returns from a carry strategy in US Treasuries, with particular regard to its well-documented business-cycle dependence.

Using the insight in Shleifer and Gennaioli (2018), we have built a simple monetary model, and we have used it to deduce how investors would form the prices of Treasury bonds under this model. We have found that, if investors were rational, no excess returns could be made under the model, and there would be no predictability of returns. However, we have shown that, if investors displayed by over-reaction (representativeness) biased described in Shleifer and Gennaioli (2018), many of the stylized features of the predictability of excess returns (such as the statistical significance of the slope factor, and the business-cyclicity of excess returns) are easily and convincingly explained. The stronger the degree of representativeness, the stronger the degree of predictability and the business-cycle dependence of the carry returns.

We have then looked at the risk-premium explanation of the excess returns, and observed that, for this explanation to hold, the business-cycle variation *of the sign* of the Sharpe ratio of the carry strategy (and hence of the risk compensation) must be linked to the correlation between the payoffs from the carry strategy and consumption. Using a variety of consumption proxies, we have failed to detect this correlation – if anything, whatever dependence we have found appeared to have the opposite sign to what required for the risk-premium explanation to hold.

Finally we have addressed a common ‘defence’ of the risk premium explanation, which points to an upward-sloping shape of the time-averaged yield curve as a *prima facie* indication that a risk premium should be at play. We have provided a simple explanation of why this is not necessarily the case – and explanation that points to the serial autocorrelation of the short-maturity rates (close to the Fed-administered Fed funds) and to the small number of major tightening / loosening cycles over the last fifty years for robust statistical conclusions to be drawn.

Our results do not ‘prove’ that risk premia do not contribute to the predictability of excess returns. However, they offer a simple and very parsimonious alternative explanation which accounts very (and arguably, more) naturally for a large number of stylized facts about excess returns.

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